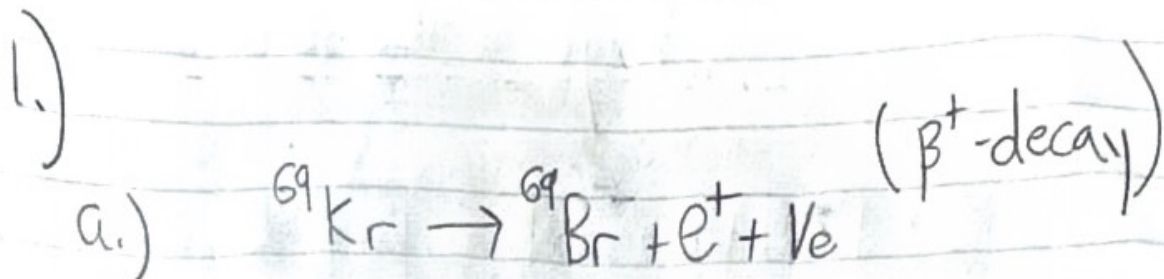


Homework #1

Parker
Lewis



Lepton Conservation test:

Left side: 0

Right side: $L_{e^+} = -1$, $L_{\nu_e} = 1$
 $L_{e^+} + L_{\nu_e} = 0$

Left side = Right side

Baryon Conservation test:

Baryon (protons, neutrons) = +1

Antibaryons = -1

Leptons (electron, positrons, etc...) = 0

36 protons + 33 neutrons $\rightarrow$ 35 protons + 34 neutrons

+69 $\rightarrow$ +69
positron + 1 neutrino

Charge Conservation Test :

Left side : charge = $36e$

Right side : charge = $35e + e = 36e$

Left side charge = Right side charge

Q-value Test :

energetically allowable :

$$Q > 0$$

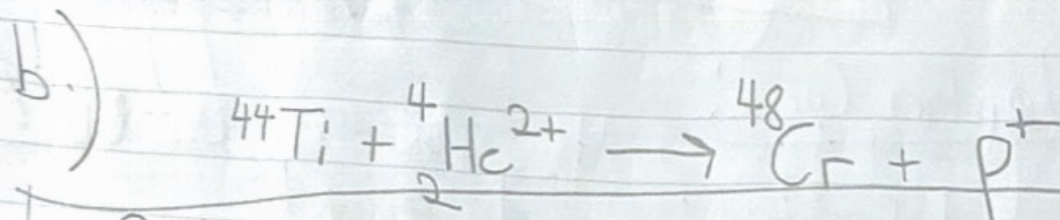
$$Q_{\beta^+} = (M(X) - M(Y) - 2m_e) c^2$$

from INDC :

$$Q_{\beta^+} = 1.31 \times 10^4 \text{ keV}$$

$$Q_{\beta^+} > 0$$

allowable reaction



fails both charge and Atomic number (baryon number) conservation.

Charge conservation:

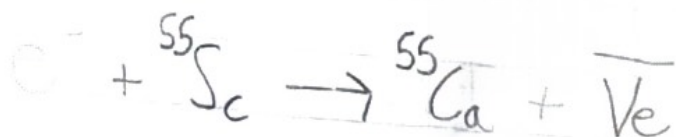
$$\text{LS} : +2 \quad \text{RS} : +1$$

Atomic number:

$$48 \neq 49$$

$$\begin{array}{l} \text{LS: } A=48 \\ \text{RS: } A=49 \end{array}$$

c.)



allowable:

Baryon Cons:

Left side: $A = 55$

Right side: $A = 55$

Left side = Right side

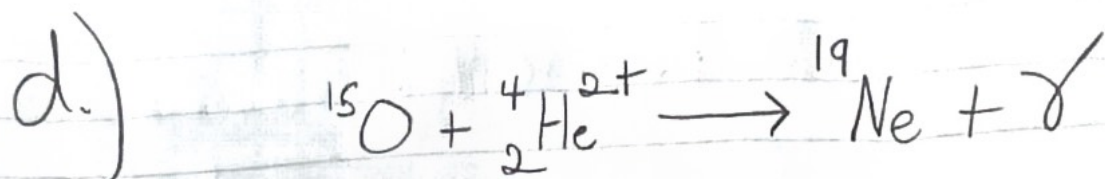
Lepton Cons:

Left side: $L = +1$

Right side: $L = -1$

$L_{\text{left}} \neq L_{\text{right}}$

Not allowable
Reaction



Atomic number (baryon number cons):

Left side = $15 + 4 = 19$

Right side = 19

Left side = Right side

Lepton number:

Left side = 0

Right side = 0

Left side = Right side

Charge conserv:

Left side: Charge = $8e + 2e = 10e$

Right side: charge = $10e$

Left side = Right side

"from nndc"

Q-value :

$$Q = 3528.5 \text{ keV}$$

$$Q > 0$$

allowable

$$n=3$$

$$2.) \quad \Delta = M(X) - A u$$

$$u = 931.494 \text{ MeV}/c^2$$

$$a.) \quad {}^{182}\text{Hf} \quad \beta^- \text{ decay}$$

$${}^{182}\text{Hf} \rightarrow {}^{182}\text{Ta} + e^- + \bar{\nu}_e$$

$$Q_{\beta^-} = \left(M(X) - M(Y) \right) c^2$$

$$= \left[\Delta(X) - \Delta(Y) \right] c^2$$

$$\Delta({}^{182}\text{Hf}) = \left[M_{\text{atom}}({}^{182}\text{Hf}) - A_{182\text{Hf}} u \right] c^2$$

* $M_{\text{atom}}({}^{182}\text{Hf})$ from nndc.

$$M_{\text{atom}}({}^{182}\text{Hf}) = 181.950564 \text{ u}$$

$$\Delta({}^{182}\text{Hf}) = \left[181.950564 - 182 \right] u c^2$$

$$= \left[-0.049436000 \right] \left(931.494 \right) \frac{\text{MeV}}{c^2} c^2$$

$$\rightarrow (\Delta(X) - \Delta(Y))c^2$$

$$\Delta(^{182}\text{Hf}) = -46.049337 \text{ MeV}$$

$$\Delta(^{182}\text{Ta}) = \text{Mass}(^{182}\text{Ta}) - Au$$

(* also will get from nndc)

$$\text{Mass}(^{182}\text{Ta}) = 181.9501546 \text{ u}$$

$$\Delta(^{182}\text{Ta}) = \left[181.9501546 - 182 \right] \times \left(931.494 \frac{\text{MeV}}{c^2} \right) c^2$$

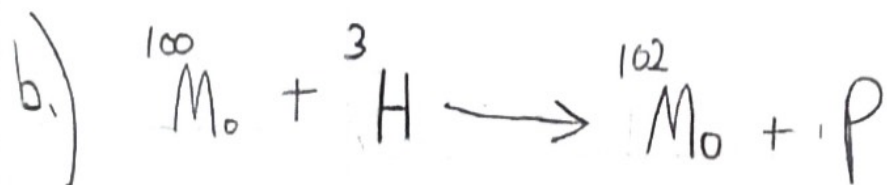
$$\Delta(^{182}\text{Ta}) = -46.430691 \text{ MeV}$$

$$(Q_{\beta^-})_{\text{calculation}} = -46.049337 + 46.430691 \text{ MeV}$$

$$= 0.38135400 \text{ MeV}$$

$$(Q_{\beta^-})_{\text{nndc}} = 381.0486 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right)$$

$$= 0.38104860$$



$$\begin{aligned} (Q)_{\text{nnde}} &= 5028.677 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) \\ &= 5.0286770 \text{ MeV} \end{aligned}$$

* All atomic masses from nnde *

$$\begin{aligned} \Delta({}^3\text{H}) &= M_{\text{atm}}({}^3\text{H}) - A u \\ &= 3.01604928132 u - 3 u \\ &= 0.016049281 u \end{aligned}$$

$$\begin{aligned} \Delta({}^{100}\text{Mo}) &= M_{\text{atm}}({}^{100}\text{Mo}) - A u \\ &= 99.9074680 u - 100 u \\ &= -0.092532000 u \end{aligned}$$

$$\begin{aligned} \Delta({}^{102}\text{Mo}) &= M_{\text{atm}}({}^{102}\text{Mo}) - A u \\ &= 101.910294 u - 102 u \\ &= -0.089706000 u \end{aligned}$$

$$\Delta(p) = 1.00782503190 u = 0.0078250319 u$$

$$(Q)_{\text{calculation}} = [0.016049281 - 0.09252000$$

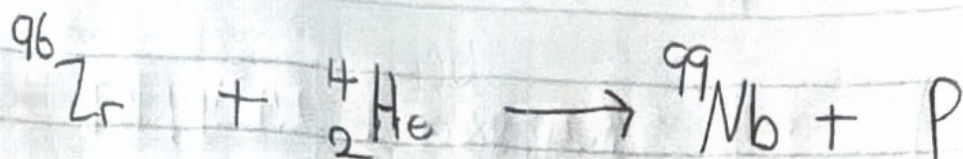
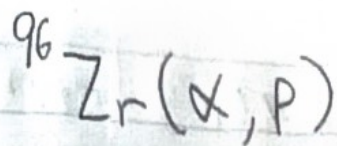
$$+ 0.09253000 - 0.089706000] \text{ u}$$

$$\left(\frac{931.494 \text{ MeV/c}^2}{1 \text{ u}} \right) \text{ c}^2$$

$$(Q)_{\text{calculation}} = 5.0284366 \text{ MeV}$$

$$(Q)_{\text{ndc}} = 5.0286770 \text{ MeV}$$

C.)



$$(Q)_{\text{calc}} = \left[\Delta(^{96}\text{Zr}) + \Delta\left({}^4_2\text{He}\right) \right] - \left[\Delta(^{99}\text{Nb}) + \Delta(p) \right]$$

• All mass excess are from nndc •

$$\Delta(^{96}\text{Zr}) = -85438.86 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = -85.438860 \text{ MeV}$$

$$\Delta\left({}^4_2\text{He}\right) = 2424.91587 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = 2.4249159 \text{ MeV}$$

$$\Delta(^{99}\text{Nb}) = -82335 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = -82.33 \text{ MeV}$$

$$\Delta(p) = 7288.97106 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = 7.2889711 \text{ MeV}$$

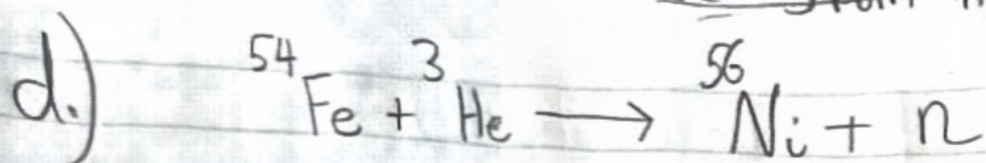
plugging in each term

$$(Q)_{\text{calc}} = \left[-85.438860 + 2.4249159 + 82.33 - 7.2889711 \right] \text{ MeV}$$

$$(Q)_{\text{calc}} = -7.9729152 \text{ MeV}$$

$$(Q)_{\text{NMDC}} = -7.9675710 \text{ MeV}$$

• Mass excesses
are from nndc



$$\Delta(^{54}\text{Fe}) = -56254.6 \text{ keV}$$

$$\Delta(n) = 8071.3181 \text{ keV}$$

$$\Delta(^3\text{He}) = 14931.21888 \text{ keV}$$

$$\Delta(^{56}\text{Ni}) = -53907.6 \text{ keV}$$

$$(Q)_{\text{calc}} = \left[-56254.6 + 14931.21888 \right] + \left[-8071.3181 + 53907.6 \right] \text{ keV}$$

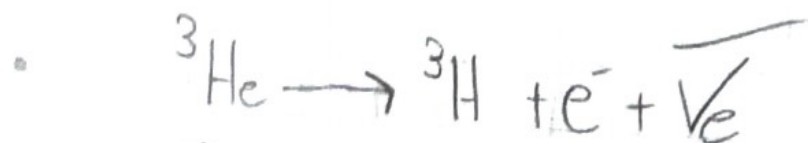
$$(Q)_{\text{calc}} = 4.5129008 \text{ MeV}$$

$$(Q)_{\text{NNDC}} = 4.5129000 \text{ MeV}$$

• Mass excesses from mDC:

3.) ${}^3\text{He}$, ${}^3\text{H}$ β^- decay

Q-value for ${}^3\text{He}$



$$Q_{\beta^-} = \left[\Delta X_{\text{parent}} - \Delta Y_{\text{daughter}} \right] \text{uc}^2$$

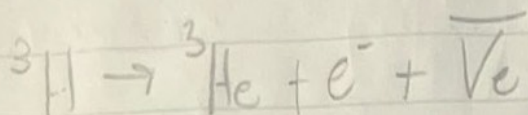
$$\Delta({}^3\text{He}) = 14931.21888 \text{ keV}$$

$$\Delta({}^3\text{H}) = 14949.81090 \text{ keV}$$

$$Q_{\beta^-} = \Delta({}^3\text{He}) - \Delta({}^3\text{H}) = -18.592020 \text{ keV}$$

${}^3\text{He}$ β^- decay forbidden because $Q_{\beta^-} < 0$

^3H beta decay!



$$Q_{\beta^-} = \Delta(^3\text{H}) - \Delta(^3\text{He})$$
$$= 18.542020 \text{ keV}$$

^3H allowed because $Q_{\beta^-} > 0$

Discussion: β^- decay occurs if
 $Q_{\beta^-} > 0$

4.)

Using nndc chart of:
nuclides

Using only chart of nuclides
is other unstable $Z=1, \dots, 81$
 $Z=43$ Tc (Technetium)

Hw #1

5.)

from nndc:
Last bound proton chlorine
isotope is ^{31}Cl , protons
are unbound starting ^{32}Cl .

$$S_p = [M(A-1, Z) + M(p) - M(A, Z)] c^2$$

$$S_p = \Delta(Z-1, N) + \Delta_p - \Delta(Z, N)$$

$\Delta = M_{\text{atom}} - A$

from
NNDc)

Chlorine-31:

$$\Delta(^{30}\text{S}) = -14059.259 \text{ keV} \left(\frac{1 \text{ MeV}}{1000} \right) = +14.059259 \text{ MeV}$$

$$\Delta(^{31}\text{Cl}) = -7034.552 \text{ keV} \left(\frac{1 \text{ MeV}}{1000} \right) = -7.034552 \text{ MeV}$$

$$\Delta(p) = 7288.971064 \text{ keV} \left(\frac{1 \text{ MeV}}{1000} \right) = 7.2889711 \text{ MeV}$$

$$S_p = (-14.059254 + 7.2889711 + 7.0345520) \text{ MeV}$$

$$= 0.26426910 \text{ MeV}$$

.) $S_p > 0$ ^{31}Cl proton bound

Chlorine -30 :

$$S_p = \Delta(^{29}\text{Si}) + \Delta(p) - \Delta(^{30}\text{Cl})$$

* (from table)

$$\Delta(p) = 7.2889711 \text{ MeV}$$

$$\Delta(^{30}\text{Cl}) = 46.745480 \text{ MeV}$$

$$\Delta(^{29}\text{Si}) = -3.0944230 \text{ MeV}$$

$$S_p = (-3.0944230 + 7.2889711 - 46.745480) \text{ MeV}$$

$$S_p \approx -0.480 \text{ MeV}$$

$S_p < 0$ ^{30}Cl
 is 1st isotope after ^{31}Cl , unbound proton.
 Proton drip line is at ^{30}Cl ,
 "starts"

6.)

(a) asymmetry term:

It's asymmetry term because
 stability is described at
 that region is $N=Z$. The asymmetry
 is what deals with contributions
 for $N \neq Z$ cases. Another way to
 think about low A nuclear landscape
 is that for nuclides (low A) to
 be stable in this region $N=Z$
 is what describes this binding energy contribution.

b)

Coulomb and asymmetry terms:

as Large A means large Z and N
 but repulsion of protons dominates
 is where also the need for stability
 also because asymmetry term comes in
 greater (most dominant effect) there
 balance is still the need for $N=Z$
 comes from why also as A is larger
 neutron richness.

c.)

Coulomb term:

Large to A/Z ratios occur
 when Coulomb repulsion dominates
 more neutrons are needed for
 stability reasons hence $A > Z$
 why we have large A/Z ratios.

d.)

Surface term and Coulomb:

high A comes from high Z
or N . High Z for Coulomb
repulsion induces higher n for
Stability purposes but too many
neutrons can cause fission.
Why not much high A nuclides.

e.)

Volume term:

Strong force binding is what
makes nuclides exist
and the volume term is what
describes binding from the
Strong force.